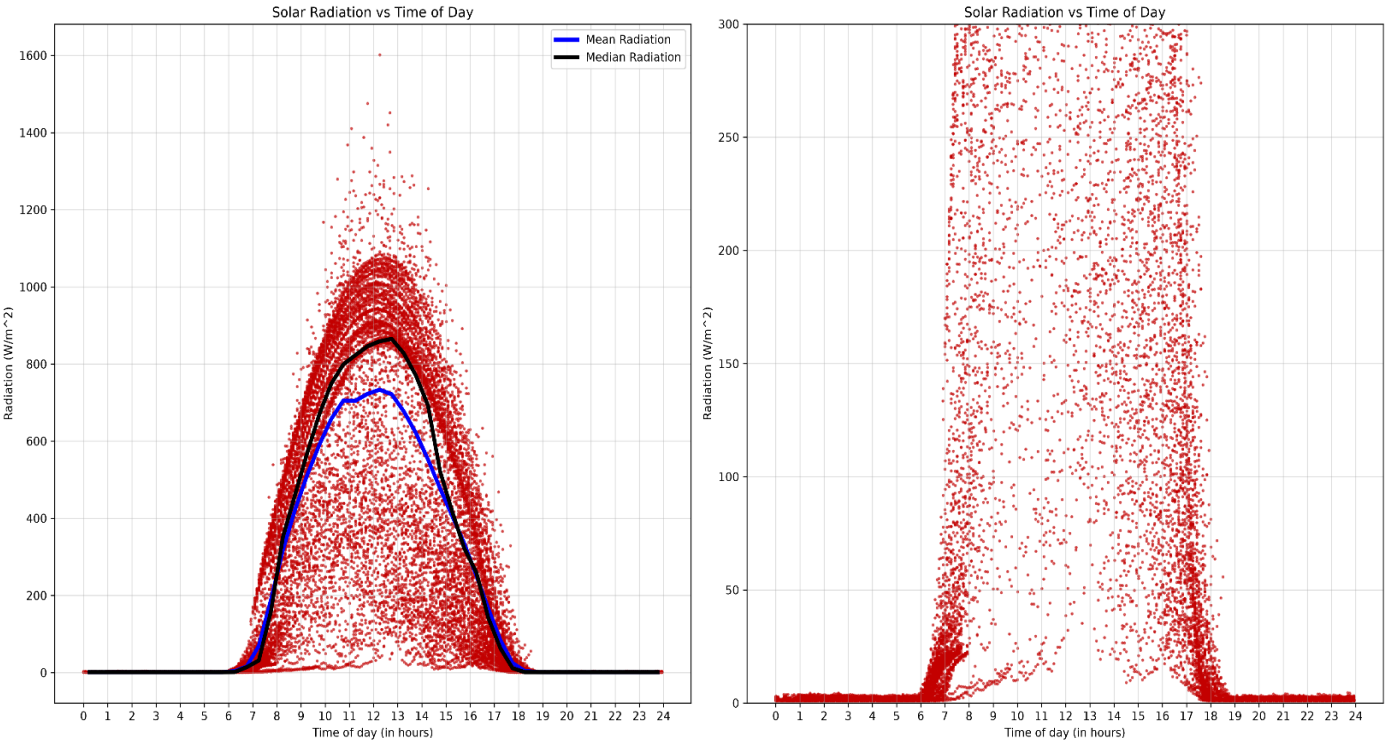
PART A

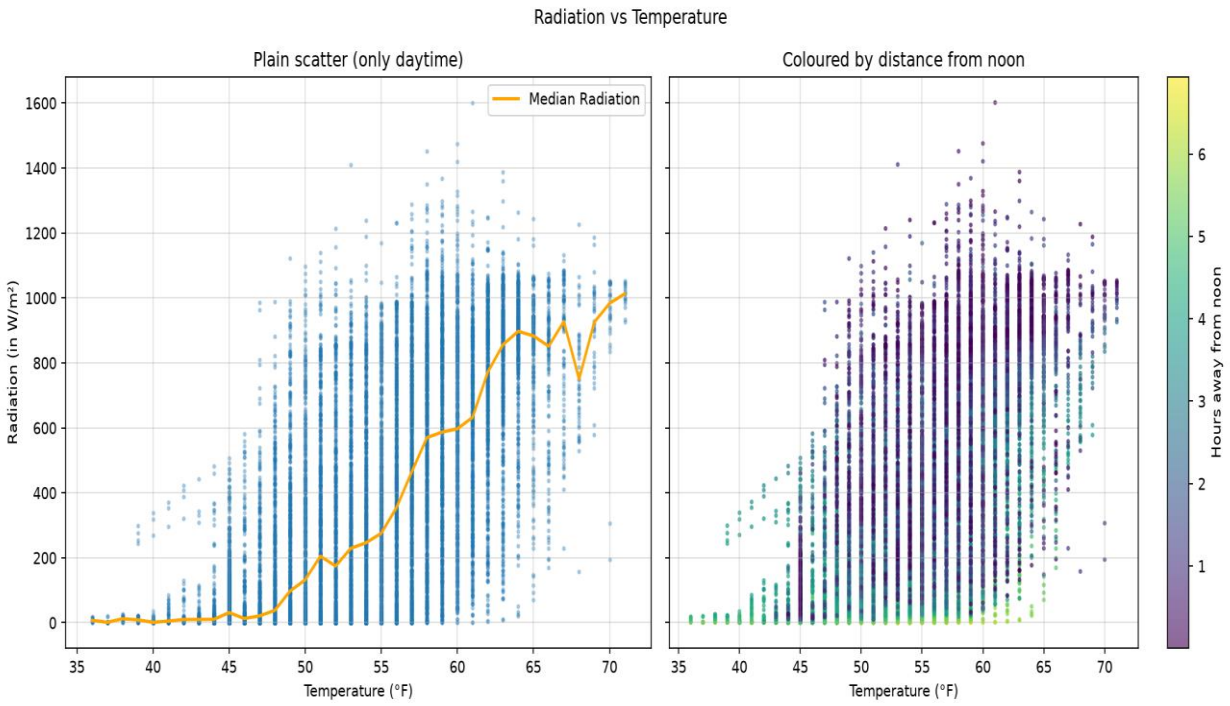
Variation of Solar Radiation vs Time of day



Fig(a)

- The above scatter plot of radiation vs time of day (in hours) gives us a visual representation of when the solar radiation peaks.
- The right graph is of a more selective range of radiation (0-300) so as to see the lower radiation values, especially during night times.

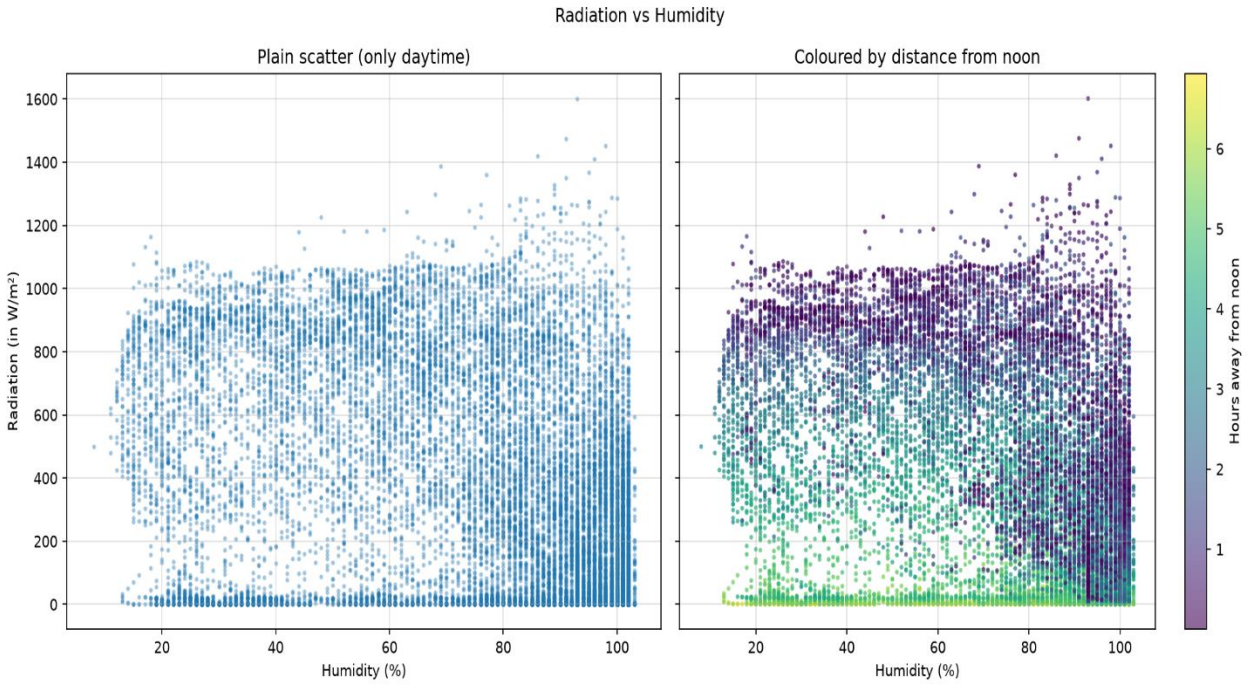
Variation of Solar Radiation vs Temperature



Fig(b)

(Graph explained in Part B)

Variation of Solar Radiation vs Humidity



Fig(c)

(Graph explained in part b)

Plots used:

a. Scatter plots to visualize how two variables are related
b. A median and mean plot was used to find a more general relationship
c. We used colour map (especially for radiation vs humidity) so that we can properly understand how humidity is related to solar radiation around noon times.
d. Also, we avoided using night times (between 7pm and 6 am) for Fig(b) and Fig(c) . During night, there is almost negligible solar radiation, due to which finding any meaningful correlation between radiation and environmental factors is out of question.

PART B

Q1. During what time of day is solar radiation generally highest?

Ans. The radiation peaks during noon, specifically between 12pm and 1pm as shown by mean curve and median curve peaks.

We can also say that the solar radiation generally peaks between sunset and sunrise. For example: on the day of the highest radiation,
peak radiation – around 12:15 pm

Sunrise- 6:08 am Sunset- 6:35pm

Clearly, the peak lies between sunset and sunrise.

Q2. Which environmental parameter appears to have the strongest relationship with radiation?

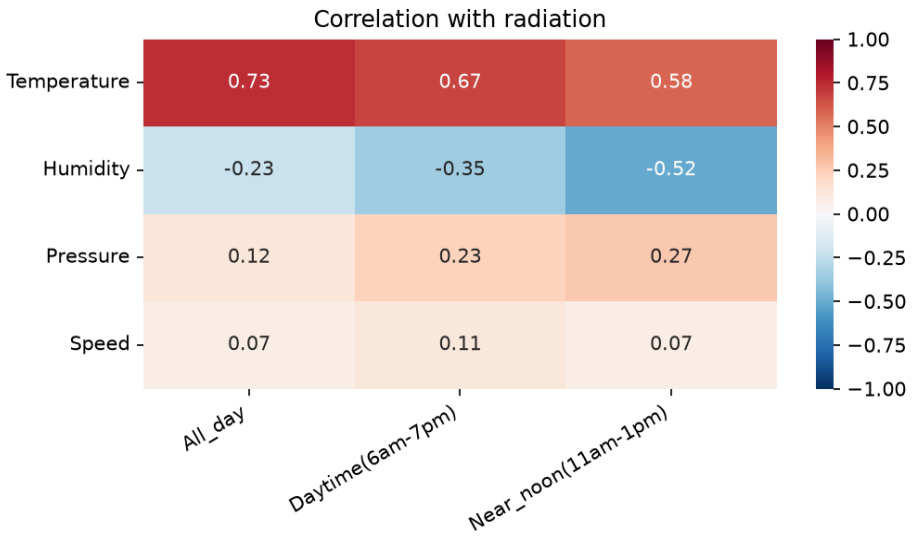
Ans. The strongest parameter definitely appears to be temperature. It has a very high positive linear correlation with Solar Radiation.

Pressure and Wind speed have little to no relation with radiation.

An interesting observation is that humidity has a negative linear correlation with Solar radiation. It is weak over full day but gets much stronger as noon approaches. The median radiations during noon time tend to be much lower for humidities in 80 to 100 percent range as compared to lower humidities.

Of course, correlation does not mean causation. Rise in radiation is accompanied with rise in temperature mainly because of daily sun-cycle.

Below is a heat map of environmental parameters in relation with radiation.



NOTE: We have not computed wind direction's correlation with radiation.

Q3. What patterns can be observed between radiation, temperature and humidity?

Ans. Temperature (Fig (b)): Radiation is almost zero when the temperature is below 44°F. This could owe to the fact that below 44°F are associated with night time and radiations are near zero at night.

As the temperature increases, radiation also increases, reaching around 900 W/m² at 65°F. Beyond this point we see fluctuations in the median because these temperatures hold on few tens of days of data as compared to hundreds of readings of other temperatures, dragging down and spiking the median plot.

Humidity (Fig(c)): In general, radiation tends to be lower when humidity is high. Around noon, radiation is highest when humidity is below 60%. At 90–100% humidity, radiation is much lower, and about 25% of the readings are below 200 W/m².

The colours show that low radiation at low humidity is mostly at sunrise and sunset (highlighted by yellow and green colours), when the sun is low. But there are also some high humidity low radiation readings around noon time (the dense purple in bottom right). Thus, humidity or other weather conditions may very well have decent influence on radiation, rather than the sun's position alone.

Q4. Are there any unusual observations or outliers? What might explain them?

Ans. Highest reading: The highest recorded radiation was 1601.26 W/m² on 4 September 2016 at 12:15 PM, when the temperature was 61°F and humidity was 93%. This is much higher than the usual noon readings, which are around 1000 W/m² or lower.

Very low-radiation days: On 18 December 2016, the mean radiation recorded was only 27.15 W/m². This could indicate very cloudy weather.

Humidity above 100%: About 13.19% of the readings showed humidity above 100%, with the highest value reaching 103%. Since relative humidity normally cannot exceed 100%, these values are likely some measurement errors.

```
Date with minimum mean radiation: 2016-12-18, Min Mean Radiation: 27.151881533101047
The max radiation on the day of minimum mean radiation is: 502.75
```

(appreciable amount of data readings)

```
Percentage of rows with humidity greater than 100%: 13.19%
Highest humidity value recorded: 103
```

PART C

Q1. Convert the Data and Time columns into a proper datetime column.

Ans.

```
                Datetime  Radiation  Temperature  Pressure  Humidity  \
0 2016-09-29 23:55:26       1.21           48     30.46        59
1 2016-09-29 23:50:23       1.21           48     30.46        58
2 2016-09-29 23:45:26       1.23           48     30.46        57
3 2016-09-29 23:40:21       1.21           48     30.46        60
4 2016-09-29 23:35:24       1.17           48     30.46        62

   WindDirection(Degrees)  Speed TimeSunRise TimeSunSet
0                  177.39   5.62    06:13:00   18:13:00
1                  176.78   3.37    06:13:00   18:13:00
2                  158.75   3.37    06:13:00   18:13:00
3                  137.71   3.37    06:13:00   18:13:00
4                  104.95   5.62    06:13:00   18:13:00
```

Q2. Determine the date with the highest recorded solar radiation.

Ans.

```
Date with maximum radiation: 2016-09-04, Max Radiation: 1601.26
```

Q3. For that day, determine the time at which maximum radiation was recorded and report the corresponding temperature, humidity and wind speed.

Ans.

```
The time at which max radiation occurs is: 12:15:04
The temperature at the time of max radiation is: 61
The humidity at the time of max radiation is: 93
The wind speed at the time of max radiation is: 9.0
```

Q4. Group observations by date and calculate the daily maximum and average radiation.

Ans.

Datetime	max	mean
2016-09-01	1073.71	298.451600
2016-09-02	1062.01	333.471219
2016-09-03	1088.09	123.974574
2016-09-04	1601.26	153.584321
2016-09-05	1264.07	306.065125
...	...	...
2016-12-27	870.52	239.910556
2016-12-28	1157.27	174.910278
2016-12-29	696.30	125.018049
2016-12-30	439.43	88.367684
2016-12-31	1026.34	144.288542

[118 rows x 2 columns]

Q5. Compare the day's maximum radiation with its sunrise and sunset times and determine how long after sunrise the peak radiation occurred.

Ans.

```
Sunrise happend at **06:08:00** and Sunset happened at **18:35:00** on the day of max radiation.
Whereas the time of max radiation was at **12:15:04**

Time taken for radiation to reach its peak from sunrise is:
6 hours, 7 minutes, and 4 seconds.
```

Q6. Identify the top 5 days with the highest average radiation.

Ans.

```
Top 5 dates with highest mean radiation:
1. Date: 2016-09-19, Mean Radiation: 335.504
2. Date: 2016-09-02, Mean Radiation: 333.471
3. Date: 2016-09-22, Mean Radiation: 325.516
4. Date: 2016-09-25, Mean Radiation: 319.922
5. Date: 2016-09-28, Mean Radiation: 312.4
```